

HDOC DAY – 6

Question 1:

A square courtyard has side `s` meters. Write a C program to find the cost of paving it if paving costs `r` rupees per square meter.

Problem understanding

- INPUT:
 1. Side of square: s meters
 2. Paving cost: r rupees per square meter
- OUTPUT:

Total cost of paving the courtyard
- FORMULA:

Area of square = $s*s$
Cost of paving = $\text{area}*r$
Therefore,
Cost = $s*s*r$

Test cases

Test case	input(s,r)	expected output	Actual output	result
1	5,30	750	750	Pass
2	10,50	5000	5000	Pass
3	60,80	288000	288000	pass

Algorithm

1. Start
2. Declare the variables for the side, paving rate, and total cost
3. Read the side 's' of the square courtyard
4. Read the paving rate 'r' per square meter
5. Calculate the area using the formula: $s*s$
6. Calculate the total paving cost by multiplying the area by 'r'
7. Display the total cost
8. Stop

c program

```
#include<stdio.h>

Int main()
{
    Float s, r, cost;
    Printf("enter the side of the square courtyard:");
    Scanf("%f", &s);
    Printf("enter the paving cost per square meter:");
    Scanf("%f", &r);

    Cost = s*s*r;
    Printf("cost of paving = %.2f rupees\n", cost);
    Return 0;
}
```

```
Session Actions Edit View Help
#include<stdio.h>
int main()
{
    float s,r, cost;
    printf("enter the side of the courtyard: ");
    scanf("%f", &s);
    printf("enter the paving cost per sqaure meter: ");
    scanf("%f", &r);

    cost = s*s*r;

    printf("cost of paving = %.2f rupees\n", cost);
    return 0;
}
```

Compiling and executing the program

```
(kali㉿kali)-[~]
$ vi hdoc6.c

(kali㉿kali)-[~]
$ gcc hdoc6.c -o hdoc6

(kali㉿kali)-[~]
$ ./hdoc6
enter the side of the courtyard: 5
enter the paving cost per sqaure meter: 30
cost of paving = 750.00 rupees

(kali㉿kali)-[~]
$
```

Evaluating the code with test cases

```
(kali㉿kali)-[~]  
$ ./hdoc6  
enter the side of the courtyard: 10  
enter the paving cost per sqaure meter: 50  
cost of paving = 5000.00 rupees  
  
(kali㉿kali)-[~]  
$ gcc hdoc6.c -o hdoc6  
  
(kali㉿kali)-[~]  
$ ./hdoc6  
enter the side of the courtyard: 60  
enter the paving cost per sqaure meter: 80  
cost of paving = 288000.00 rupees
```

The expected output matches the actual output hence the code and calculations are correct.

Question 2:

Write a C program to find the area and perimeter of a kite when its two diagonals and two unequal adjacent sides are given.

Problem understanding

- INPUT:
 1. First diagonal = d1
 2. Second diagonal = d2
 3. First adjacent side = a
 4. Second adjacent side = b
- OUTPUT:
 1. Area of the kite
 2. Perimeter of the kite
- FORMULA:

Area of kite: $(d1*d2)/2$

Since the two pairs of adjacent sides are equal:

$$\text{Perimeter} = 2*(a+b)$$

Test cases

Test case	Input (d1, d2, a, b)	Expected output	Actual output	result
1	d1=5 d2=7 a=9 b=10	Area=17.50 perimeter=38.00	Area=17.50 perimeter=38.00	Pass
2	d1=12 d2=10 a=7 b=5	Area=60 perimeter=24	Area=60 perimeter=24	Pass
3	d1=20 d2=15 a=10 b=8	Area=150 perimeter=36	Area=150 perimeter=36	Pass

Algorithm

1. Start
2. Declare variables for the two diagonals and the two unequal adjacent sides
3. Read the first diagonal d1
4. Read the second diagonal d2
5. Read the first adjacent side a
6. Read the second adjacent side b
7. Calculate the area using the formula: $(d1*d2) / 2$
8. Calculate the perimeter using the formula: $2*(a+b)$
9. Display the area and perimeter
10. Stop

C program

```

#include<stdio.h>

Int main()
{
    Float d1, d2, a, b, area, perimeter;

    Printf("enter the length of first diagonal:");

    Scanf("%f", &d1);

    Printf("enter the length of second diagonal:");

    Scanf("%f", &d2);

    Printf("enter the length of first adjacent side:");

    Scanf("%f", &a);

    Printf("enter the length of second adjacent side:");

    Scanf("%f", &b);

    Area = (d1*d2)/2

    Perimeter = 2*(a+b)

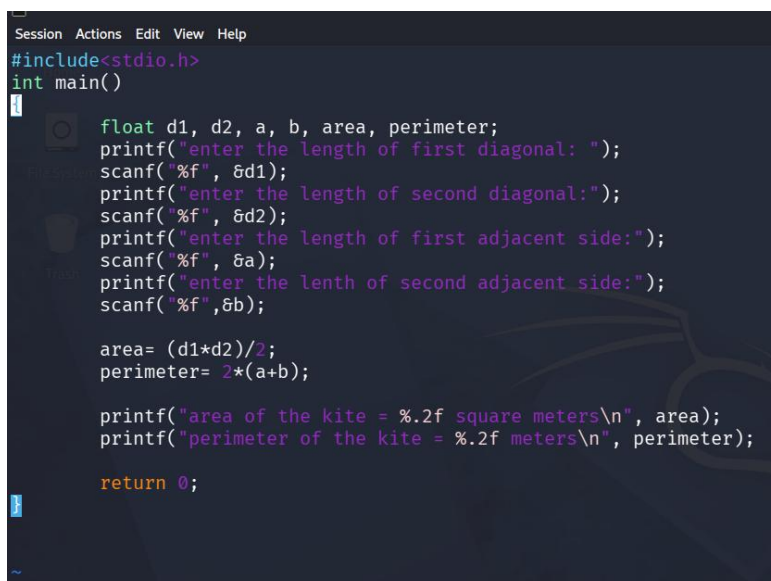
    Printf("area of the kite = %.2f square meters\n", area);

    Printf("perimeter of the kite = %.2f square meters\n", perimeter);

    return 0;

}

```



```

Session Actions Edit View Help
#include<stdio.h>
int main()
{
    float d1, d2, a, b, area, perimeter;
    printf("enter the length of first diagonal: ");
    scanf("%f", &d1);
    printf("enter the length of second diagonal:");
    scanf("%f", &d2);
    printf("enter the length of first adjacent side:");
    scanf("%f", &a);
    printf("enter the length of second adjacent side:");
    scanf("%f", &b);

    area= (d1*d2)/2;
    perimeter= 2*(a+b);

    printf("area of the kite = %.2f square meters\n", area);
    printf("perimeter of the kite = %.2f meters\n", perimeter);

    return 0;
}

```

Compiling and executing the program

```
(kali㉿kali)-[~]  
$ vi kite.c  
  
(kali㉿kali)-[~]  
$ gcc kite.c -o kite  
  
(kali㉿kali)-[~]  
$ ./kite  
enter the length of first diagonal: 5  
enter the length of second diagonal:7  
enter the length of first adjacent side:9  
enter the length of second adjacent side:10  
area of the kite = 17.50 square meters  
perimeter of the kite = 38.00 meters  
  
(kali㉿kali)-[~]  
$
```

Evaluating the code with test cases

```
(kali㉿kali)-[~]  
$ ./kite  
enter the length of first diagonal: 12  
enter the length of second diagonal:10  
enter the length of first adjacent side:7  
enter the length of second adjacent side:5  
area of the kite = 60.00 square meters  
perimeter of the kite = 24.00 meters  
  
(kali㉿kali)-[~]  
$ ./kite  
enter the length of first diagonal: 20  
enter the length of second diagonal:15  
enter the length of first adjacent side:10  
enter the length of second adjacent side:8  
area of the kite = 150.00 square meters  
perimeter of the kite = 36.00 meters
```

